

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460378.7197 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.019
Transit depth (Rp/Rs)^2	1.97 +/- 0.54 [%]
Orbital Inclination (inc)	83.21 +/- 1.34
Airmass coefficient 1 (a1)	1.257 +/- 0.014
Airmass coefficient 2 (a2)	-0.159 +/- 0.036
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	None
Optimal Aperture	3.78
Optimal Annulus	14.18
Transit Duration (day)	0.075 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.019 vs 0.1368 (deviation 0.17σ)
Duration fit vs published	0.075 d vs 0.0758 d (deviation 0.08σ)
Mid-time O–C	3.2 min
Depth SNR	7.4
Residual scatter	1.14 %

Light curve

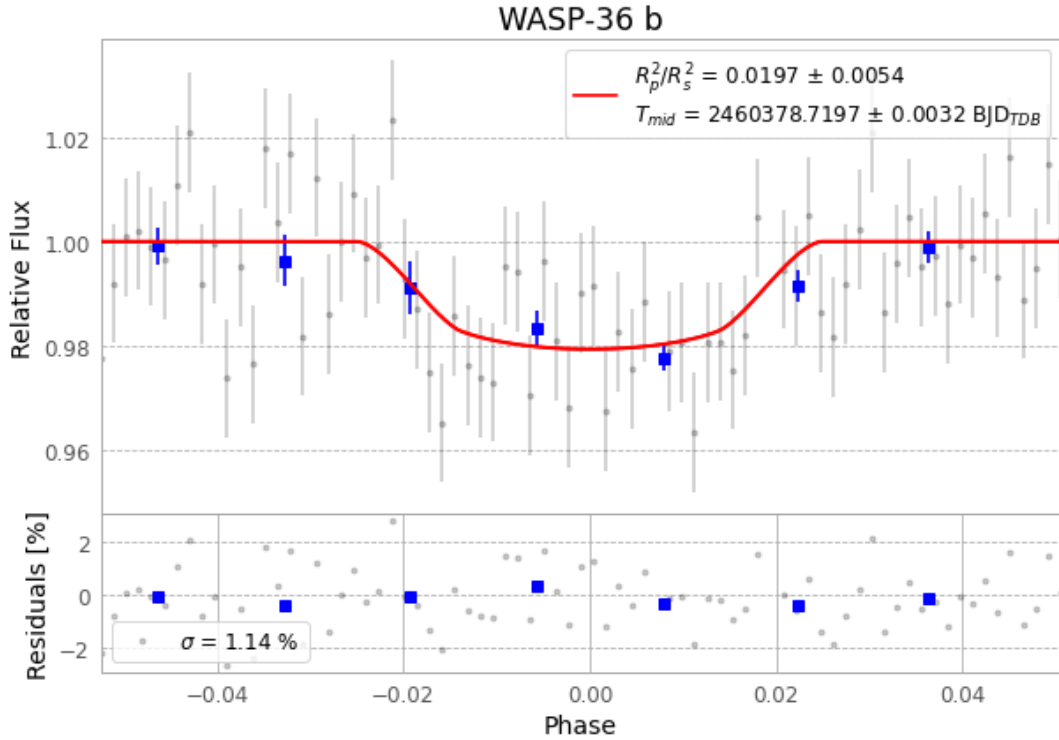


Figure 1 — FinalLightCurve_WASP-36 b_2024-03-08.png (SHA-256 e264c8185f765f6f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fe95c61b5d20193b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36240309031215.FITS → WASP-36240309070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240309.log (8753e8666788...), FinalLightCurve_WASP-36 b_2024-03-08.png (e264c8185f76...), FinalLightCurve_WASP-36 b_2024-03-08.csv (2031fef97e58...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 08:12Z.